

George Ma

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EDUCATION

University of California, San Diego

La Jolla, CA

B.S. in Computer Science; GPA: 3.94

Expected June 2028

Relevant Courses: Design & Analysis of Algorithms, Advanced Data Structures, Data Structures & Algorithms, Systems Programming, Database Systems, Object-Oriented Programming, Computer Organization, Data Science Theory

TECHNICAL SKILLS

Languages: Go, Python, Java, C/C++, SQL, TypeScript, JavaScript, Bash/Shell, HTML, CSS, ARM Assembly

Libraries/Frameworks: FastAPI, Flask, Node.js, Express.js, LiteLLM, PyTorch, scikit-learn, React, React Native

Cloud/Infra: AWS, GCP, Kubernetes, Docker/Podman, Terraform, Jenkins, ArgoCD, Kafka, PostgreSQL/MySQL, Git

EXPERIENCE

Capital One

McLean, VA

Software Engineer Intern – Enterprise Platforms (DevOps/SRE)

June 2026 – August 2026

- Built **PR-triggered** Claude skills to translate **Gatling/JMeter** scripts and dynamically generate **k6 test suites** from OpenAPI specs and load parameters, cutting API performance test modernization time by **70%** for **5k users**
- Deployed an automated test execution layer using ephemeral **AWS ECS** instances via **Jenkins CI** pipelines to support distributed load generation for high-traffic APIs, scaling enterprise performance testing reliability by **67%**
- Engineered an event-driven **Spring Boot notification service** for test statuses using a **Kafka** message broker and **Redis** for SMTP rate-limiting, mitigating spam via a Vue.js and PostgreSQL user preference dashboard

IBM

San Jose, CA

Software Engineer Intern – Automation and AI (Full-Stack & Infrastructure)

February 2026 – May 2026

- Built a React Native/Expo mobile AI assistant, optimizing chat rendering using FlatList for **7k+ internal users**
- Engineered an **OAuth/OIDC** gateway in **Go** via PKCE, JWT/Redis-backed sessions, and **OPA-based RBAC**
- Co-designed a FastAPI-based **LiteLLM proxy** to route custom TypeScript agent workloads across AWS Bedrock, IBM WatsonX, and Mistral AI, reducing chat inference pipeline latency by **25%** through semantic model routing
- Automated **OpenShift (K8s)** deployment for backend, agent, and auth microservices using Terraform/Jenkins
- Optimized Strimzi-managed **Kafka** pipeline latency by **32%** by diagnosing bottlenecks with Prometheus/Grafana

Praxie AI

San Francisco, CA

Software Engineer Intern – Product & Platform

April 2025 – February 2026

- Migrated backend infra from Firebase to a containerized Express REST API on **Google Cloud Run**, provisioning services with **Terraform** and securing GitHub Actions CI/CD pipelines using Workload Identity Federation
- Shipped scalable tournament and multi-player management features in React Native using **PostGIS** geo-indexing and TanStack Query for optimized client-side caching, achieving **sub-1s search latency** for 300+ active users
- Developed a fault-tolerant **NoSQL to PostgreSQL** (CloudSQL) migration pipeline by using node-pg-migrate to refactor schemas and backfill production data, improving query performance by **20%** through denormalization

PROJECTS

Neural eBPF | Go, Cilium/eBPF, Linux, Kubernetes, LangGraph, Redis, React

- Built a **self-healing** SRE system using a Go daemon to detect CPU/FD leaks via eBPF `sched_switch` tracepoints
- Engineered a Redis-backed **LangGraph agent** to automate userspace mitigations, cutting recovery to **under 10s**
- Deployed a Minikube HTTP/SSE observability stack via cgroup/debugfs mounts, reducing manual triage by **70%**

Shadow Infra | Go, FastAPI, AWS, Kubernetes, Docker, LangGraph, Supabase, React

- Architected a **PR-triggered shadow deployment** testing system using FastAPI/GitHub webhooks for automated creation/teardown of ephemeral **AWS EKS** services, catching **87%** of seeded API regressions before merging PRs
- Engineered an async **Go tee proxy** to mirror traffic to production and shadow services with no user-facing delay
- Optimized a LangGraph drift-analysis pipeline with fast-path checks, cutting manual regression review by **80%**

Settle | Kotlin, Jetpack Compose, CameraX, ML Kit, LiteRT-LM, Gemma, Qualcomm NPU

- Built an Android **legal-doc scanner** with CameraX/ML Kit OCR, mapping clauses to coordinate-based boxes
- Deployed Gemma 4 E2B via **LiteRT-LM** and Snapdragon **Hexagon NPU** for fully local text classification/Q&A
- Implemented 2-tier risk labels, bottom-sheet rationale, and streamed follow-up chat via **Kotlin coroutines/Flow**